

Process Variables Measurement

□ To impart students with the fundamental concepts of pressure measurement and its

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4081 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4081.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	4081
Course title	Process Variables Measurement
Semester	4 Credits: 4
Course category	Program Core

Item	Official information
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

12 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart students with the fundamental concepts of pressure measurement and its applications in industries.
- To explain the construction, working principle and applications of different types of flow meters.
- To give an insight about different techniques and applications of level measurement.
- To illustrate the fundamental concepts and applications of different types of temperature measuring techniques.

Course outcomes

CO	Official outcome	Study level
CO1	applications of different types of pressure 15 Applying measuring instruments. Explain the construction, working principle and	See official syllabus
CO2	15 Understanding applications of different types of flow meters. Explain the different techniques for level	See official syllabus
CO3	12 Understanding measurement. Apply different types of transducers for	See official syllabus
CO4	16 Applying temperature measurement	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
C01	C01 3
C02	C02 2
C03	C03 2

Row	Extracted official mapping
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	different types of pressure measuring instruments.	6	As specified
Module 2	types of flow meters.	2	As specified
Module 3	Explain the different techniques for level measurement.	2	As specified
Module 4	Apply different types of transducers for temperature measurement	2	As specified

MODULE 1

different types of pressure measuring instruments.

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.

- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: different types of pressure measuring instruments.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Applying standards. Describe the construction and working

2 Applying standards. Describe the construction and working is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying standards. Describe the construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying standards. describe the construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying standards. Describe the construction and working means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-2 Applying standards. Describe the construction and working of a standard. definition/meaning of a standard. steps, application of a standard. Technical terms English- Malayalam.

3 Understanding principle of different types of manometers Explain the construction and working

3 Understanding principle of different types of manometers Explain the construction and working is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding principle of different types of manometers Explain the construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding principle of different types of manometers explain the construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding principle of different types of manometers Explain the construction and working means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding principle of different types of manometers Explain the construction and working of different types of manometers. definition/meaning of manometer. steps, application of manometer. Technical terms English-Malayalam.

3 Understanding principle of elastic pressure gauges Compare and apply different transducers for

3 Understanding principle of elastic pressure gauges Compare and apply different transducers for is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding principle of elastic pressure gauges compare and apply different transducers for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding principle of elastic pressure gauges Compare and apply different transducers for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding principle of elastic pressure gauges Compare and apply different transducers for ഏകദേശം അളക്കൽ definition/meaning അളക്കൽ. അളക്കൽ അളക്കൽ, steps, application അളക്കൽ അളക്കൽ അളക്കൽ. Technical terms English-എ അളക്കൽ അളക്കൽ.

3 Applying various pressure measurements. Explain different methods used for vacuum

3 Applying various pressure measurements. Explain different methods used for vacuum is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying various pressure measurements. Explain different methods used for vacuum using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying various pressure measurements. explain different methods used for vacuum should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying various pressure measurements. Explain different methods used for vacuum means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Applying various pressure measurements. Explain different methods used for vacuum പരീക്ഷണം. definition/meaning പരീക്ഷണം. പരീക്ഷണം പരീക്ഷണം, steps, application പരീക്ഷണം പരീക്ഷണം പരീക്ഷണം. Technical terms English-പരീക്ഷണം പരീക്ഷണം.

2 Understanding measurement Calibrate various pressure measuring

2 Understanding measurement Calibrate various pressure measuring is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding measurement Calibrate various pressure measuring using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding measurement calibrate various pressure measuring should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding measurement Calibrate various pressure measuring means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-2 Understanding measurement Calibrate various pressure measuring instruments definition/meaning ഉപയോഗം. ഉപയോഗം ഉപയോഗം, steps, application ഉപയോഗം ഉപയോഗം ഉപയോഗം. Technical terms English-ഉപയോഗം ഉപയോഗം.

2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure - simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows - Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers -pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different

2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure - simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows - Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter - Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers -pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different is an official topic in Module 1 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure - simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows - Diaphragms-Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo

electric pressure sensor-pressure switches-- Differential pressure transmitter - Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers - pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding instruments. definition of pressure - different types of pressure - units of pressure – simple conversion problems-methods of pressure measurement- manometers- u tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-elastic pressure transducers-c-type and spiral bourdon gauges - bellows – diaphragms-electrical pressure transducers-strain gauges- capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- differential pressure transmitter -vacuum gauges- mcleod gauge, pirani gauge - ionization gauge -testing and calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. i/ p and p/i converters using flapper nozzle system. explain the construction, working principle and applications of different should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding instruments. Definition of pressure - different types of pressure - units of pressure – simple conversion problems-Methods of pressure measurement- manometers- U tube manometers, well type manometers, inclined manometers, errors in manometers- simple problems-Elastic pressure transducers-C-type and spiral bourdon gauges - Bellows – Diaphragms- Electrical pressure transducers- Strain gauges- Capacitive type pressure gauges- piezo electric pressure sensor-pressure switches-- Differential pressure transmitter -Vacuum gauges- McLeod gauge, Pirani gauge - Ionization gauge -Testing and Calibration of pressure measuring instruments - dead weight tester- manometers –pressure transmitters. I/ P and P/I converters using flapper nozzle system. Explain the construction, working principle and applications of different means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

different types of pressure measuring instruments.		
Convert between different pressure		
M1.01		2
Applying		
standards.		
Describe the construction and working		
M1.02		3
Understanding		
principle of different types of manometers		
Explain the construction and working		
M1.03		3
Understanding		
principle of elastic pressure gauges		
Compare and apply different transducers for		
M1.04		3
Applying		
various pressure measurements.		
Explain different methods used for vacuum		
M1.05		2
Understanding		
measurement		
Calibrate various pressure measuring		
M1.06		2
Understanding		
instruments.		
Contents:		
Definition of pressure - different types of pressure - units of pressure – simple		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

types of flow meters.

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: types of flow meters.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring

4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring is an official topic in Module 2 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding measurement. compare the construction and working principle of different types of flow 8 understanding m2.02 measurement techniques. calibrate various flow measuring should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding measurement. Compare the construction and working principle of different types of flow measurement techniques. Calibrate various flow measuring devices. definition/meaning. steps, application. Technical terms English- Malayalam.

3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow - Reynold's number - Continuity equation - Bernoulli's equation - simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle - Pitot tube - simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters -dilution gauging/ tracer method- gravimetric method.

3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow - Reynold's number - Continuity equation - Bernoulli's equation - simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle - Pitot tube - simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters -dilution gauging/ tracer method- gravimetric method. is an official topic in Module 2 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow - Reynold's number - Continuity equation - Bernoulli's equation - simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle - Pitot tube - simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters -dilution gauging/ tracer method- gravimetric method. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding instruments. principles of flow measurements- laminar and turbulent flow – reynold's number - continuity equation - bernoulli's equation – simple problems-methods of flow measurement- variable head flow meter- orifice plate flow meter - venturi meter - flow nozzle – pitot tube – simple problems- positive displacement flow meter - nutating disc -. variable area flow meter - rotameter- coriolis type mass flow meter. electromagnetic flow meters. - ultrasonic flow meters- calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli's equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow – Reynold's number - Continuity equation - Bernoulli's equation – simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

types of flow meters.

Recall the fundamentals of flow

M2.01

4

Understanding

measurement.

Compare the construction and working
principle of different types of flow

8

Understanding

M2.02

measurement techniques.

Calibrate various flow measuring

M2.03

3

Understanding

instruments.

Series Test - I

1

Contents:

Principles of Flow measurements- Laminar and turbulent flow – Reynold's number -
Continuity equation - Bernoulli's equation – simple problems-Methods of flow
measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter -
Flow

Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating
disc -.

Variable area flow meter - Rotameter- Coriolis type mass flow meter.

Electromagnetic

flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution
gauging/

Module study routine: Read the source wording, make a one-page concept map,
practise one short answer and one structured answer, then review the Malayalam
support notes.

MODULE 3

Explain the different techniques for level measurement.

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the different techniques for level measurement.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers

principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers is an official topic in Module 3 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, principle of different types of level 8 understanding measurement techniques. compare and apply different transducers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- principle of different types of level 8
Understanding measurement techniques. Compare and apply different transducers
definition/meaning . steps, application . Technical terms English-
.

for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type - hydro static pressure type- . Air purge type level indicator - electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge - microwave Level switches - Differential Pressure type level transmitter - simple problems.

for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type - hydro static pressure type- . Air purge type level indicator - electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge - microwave Level switches - Differential Pressure type level transmitter - simple problems. is an official topic in Module 3 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type - hydro static pressure type- . Air purge type level indicator - electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge - microwave Level switches - Differential Pressure type level transmitter - simple problems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, for various level measurements. methods of liquid level measurement- sight glass technique - float type - displacer and torque tube type - hydro static pressure type- . air purge type level indicator - electrical methods- capacitive level indicator- radiation level detector - ultrasonic level gauge - microwave level switches - differential pressure type level transmitter - simple problems. should be

discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Explain the different techniques for level measurement.

M3.01	Explain the construction and working principle of different types of level measurement techniques.	8
Understanding	Compare and apply different transducers	

Understanding

M3.02 for various level measurements.

4

Contents:

Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type – hydro static pressure type- . Air purge type level indicator – electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge – microwave Level switches – Differential Pressure type level transmitter – simple problems.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Apply different types of transducers for temperature measurement

This module is presented from the official syllabus scope for Process Variables Measurement. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply different types of transducers for temperature measurement
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying

Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying is an official topic in Module 4 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, convert different temperature scales. 3 applying explain the construction and working principle of different types of temperature 8 understanding m4.02 measurement techniques compare and apply different transducers applying should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying means in the official course context

5 for various temperature measurement Definition of temperature - different temperature scales and conversions - simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer- Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC - construction and working of Resistance Temperature Detector - different types (Ni, Cu, & Pt) - characteristics of RTD - simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect - laws of thermocouple - law of intermediate temperature and law of intermediate metals - different types of industrial thermocouples- E,J,T,K,S,R - properties of different types- simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link

1 <https://www.youtube.com/watch?v=4knAR4HzRJs> 2 www.temperatures.com

› Vendors › Electronics › A To D 3 <https://www.youtube.com/watch?v=As5kzxkyT24> 4 <https://www.youtube.com/watch?v=K2CH1cUkMgs> 5 <https://www.youtube.com/watch?v=JHCS7ldf4aU> 6 [nptel.ac.in/courses/101106040/chapter 5.pdf](http://nptel.ac.in/courses/101106040/chapter%205.pdf) 7 <https://www.youtube.com/watch?v=-MQU0xgh6bA> 8 <https://instrumentationtools.com/dp-transmitters-level-measurement> 9 <https://www.emerson.com/.../measurement-instrumentation/level>

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Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=4knAR4HzRJs>

2 www.temperatures.com > Vendors > Electronics > A To D 3

<https://www.youtube.com/watch?v=As5kzxkyT24> 4

<https://www.youtube.com/watch?v=K2CH1cUkMgs> 5

<https://www.youtube.com/watch?v=JHCS7ldf4aU> 6

[nptel.ac.in/courses/101106040/chapter 5.pdf](http://nptel.ac.in/courses/101106040/chapter%205.pdf) 7 [https://www.youtube.com/watch?](https://www.youtube.com/watch?v=-MQU0xgh6bA)

[v=-MQU0xgh6bA](https://www.youtube.com/watch?v=-MQU0xgh6bA) 8 [https://instrumentationtools.com/dp-transmitters-level-](https://instrumentationtools.com/dp-transmitters-level-measurement)

[measurement](https://www.emerson.com/.../measurement-instrumentation/level) 9 <https://www.emerson.com/.../measurement-instrumentation/level>

is an official topic in Module 4 of Process Variables Measurement. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types (Ni, Cu, & Pt) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types- simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=4knAR4HzRJs> 2 www.temperatures.com > Vendors > Electronics > A To D 3 <https://www.youtube.com/watch?v=As5kzxkyT24> 4 <https://www.youtube.com/watch?v=K2CH1cUkMgs> 5 <https://www.youtube.com/watch?v=JHCS7ldf4aU> 6 [nptel.ac.in/courses/101106040/chapter 5.pdf](http://nptel.ac.in/courses/101106040/chapter%205.pdf) 7 <https://www.youtube.com/watch?v=-MQU0xgh6bA> 8 <https://instrumentationtools.com/dp-transmitters-level->

measurement 9 <https://www.emerson.com/.../measurement-instrumentation/level> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 for various temperature measurement definition of temperature - different temperature scales and conversions – simple problems. working of bimetallic thermometer, mercury in glass thermometer, mercury in steel thermometer-construction and working of radiation pyrometer, optical pyrometer-definition of ptc and ntc – construction and working of resistance temperature detector – different types (ni, cu, & pt) - characteristics of rtd – simple problems - thermistor and its types- properties- thermistor characteristics- applications of thermistor-seebeck effect - peltier effect and thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- e,j,t,k,s,r - properties of different types- simple problems- thermopile - compare the characteristics of thermocouple, rtd and thermistor. text / reference: t/r book title/author a.k. sawhney, a course in mechanical measurements and instrumentation, r1 dhanpat rai r2 s.k singh , industrial instrumentation and control , tmh r3 r.k jain , mechanical and industrial measurements , khanna publishers r4 krishnaswami , industrial instrumentation , new age publications online resources: sl.no website link 1 <https://www.youtube.com/watch?v=4knar4hzrjs> 2 [www.temperatures.com > vendors > electronics > a to d](https://www.temperatures.com/vendors/electronics/a-to-d) 3 <https://www.youtube.com/watch?v=as5kzxkyt24> 4 <https://www.youtube.com/watch?v=k2ch1cukmgs> 5 <https://www.youtube.com/watch?v=jhcs7idf4au> 6 [nptel.ac.in/courses/101106040/chapter 5.pdf](https://nptel.ac.in/courses/101106040/chapter%205.pdf) 7 <https://www.youtube.com/watch?v=-mqu0xgh6ba> 8 <https://instrumentationtools.com/dp-transmitters-level-measurement> 9 <https://www.emerson.com/.../measurement-instrumentation/level> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 for various temperature measurement Definition of temperature - different temperature scales and conversions – simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types (Ni, Cu, & Pt) - characteristics of RTD – simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor- Seebeck effect - Peltier effect and Thomson effect – laws of thermocouple – law of intermediate temperature and law of intermediate metals – different types of industrial thermocouples- E,J,T,K,S,R - properties of different types- simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K.

Aspect	What to prepare
	Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=4knAR4HzRJs 2 www.temperatures.com > Vendors > Electronics > A To D 3 https://www.youtube.com/watch?v=As5kzxkyT24 4 https://www.youtube.com/watch?v=K2CH1cUkMgs 5 https://www.youtube.com/watch?v=JHCS7ldf4aU 6 nptel.ac.in/courses/101106040/chapter 5.pdf 7 https://www.youtube.com/watch?v=-MQU0xgh6bA 8 https://instrumentationtools.com/dp-transmitters-level-measurement 9 https://www.emerson.com/.../measurement-instrumentation/level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-5 for various temperature measurement
Definition of temperature - different temperature scales and conversions - simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC - construction and working of Resistance Temperature Detector - different types (Ni, Cu, & Pt) - characteristics of RTD - simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect - laws of thermocouple - law of intermediate temperature and law of intermediate metals - different types of industrial thermocouples- E,J,T,K,S,R - properties of different types- simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=4knAR4HzRJs> 2 www.temperatures.com > Vendors > Electronics > A To D 3 <https://www.youtube.com/watch?v=As5kzxkyT24> 4 <https://www.youtube.com/watch?v=K2CH1cUkMgs> 5 <https://www.youtube.com/watch?v=JHCS7ldf4aU> 6 [nptel.ac.in/courses/101106040/chapter 5.pdf](http://nptel.ac.in/courses/101106040/chapter%205.pdf) 7 <https://www.youtube.com/watch?v=-MQU0xgh6bA> 8 <https://instrumentationtools.com/dp-transmitters-level-measurement> 9 <https://www.emerson.com/.../measurement-instrumentation/level> means in the official course context

v=-MQU0xgh6bA 8 <https://instrumentationtools.com/dp-transmitters-level-measurement> 9 <https://www.emerson.com/.../measurement-instrumentation/level> definition/meaning steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Apply different types of transducers for temperature measurement

M4.01 Convert different temperature scales. 3
Applying

Explain the construction and working principle of different types of temperature measurement techniques 8
Understanding

M4.02 measurement techniques
Compare and apply different transducers

Applying
M4.03 5

for various temperature measurement
Series Test - I 1

Contents:

Definition of temperature - different temperature scales and conversions – simple problems.

Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel

thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC – construction and working of Resistance Temperature Detector – different types (Ni, Cu, & Pt) - characteristics of RTD – simple problems -

Thermistor and its types- properties- Thermistor Characteristics- Applications of

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Applying standards. Describe the construction and working. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Applying standards*. Describe the construction and working. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding principle of different types of manometers
Explain the construction and working. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding principle of different types of manometers* Explain the construction and working. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding principle of elastic pressure gauges
Compare and apply different transducers for. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding principle of elastic pressure gauges* Compare and apply different transducers for. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying various pressure measurements. Explain different methods used for vacuum. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Applying various pressure measurements. Explain different methods used for vacuum.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow - Reynold's number - Continuity equation - Bernoulli's equation - simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow Nozzle - Pitot tube - simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters- Calibration of liquid flow meters -dilution gauging/ tracer method- gravimetric method.. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding instruments. Principles of Flow measurements- Laminar and turbulent flow - Reynold's number - Continuity equation - Bernoulli's equation - simple problems-Methods of flow measurement- Variable Head flow meter- Orifice plate flow meter - Venturi meter - Flow*

Nozzle – Pitot tube – simple problems-Positive displacement flow meter - Nutating disc -. Variable area flow meter - Rotameter- Coriolis type mass flow meter. Electromagnetic flow meters. - Ultrasonic flow meters-Calibration of liquid flow meters –dilution gauging/ tracer method- gravimetric method.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers. (3 marks)

Answer: Begin with the standard definition or identity of *principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type - hydro static pressure type- . Air purge type level indicator - electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge - microwave Level switches - Differential Pressure type level transmitter - simple problems.. (3 marks)

Answer: Begin with the standard definition or identity of *for various level measurements. Methods of liquid level measurement- Sight glass technique - Float type - Displacer and torque tube type - hydro static pressure type- . Air purge type level indicator - electrical methods- Capacitive level indicator- radiation level detector - Ultrasonic level gauge - microwave Level switches - Differential Pressure type level transmitter - simple problems..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying. (3 marks)

Answer: Begin with the standard definition or identity of *Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 5 for various temperature measurement Definition of temperature - different temperature scales and conversions - simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC - construction and working of Resistance Temperature Detector - different types (Ni, Cu, & Pt) - characteristics of RTD - simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect - laws of thermocouple - law of intermediate temperature and law of intermediate metals - different types of industrial thermocouples- E,J,T,K,S,R - properties of different types- simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=4knAR4HzRJs> 2 www.temperatures.com > Vendors > Electronics > A To D 3 <https://www.youtube.com/watch?>

v=As5kzxkyT24 4 <https://www.youtube.com/watch?v=K2CH1cUkMgs> 5
<https://www.youtube.com/watch?v=JHCS7ldf4aU> 6
<https://www.youtube.com/watch?v=-MQU0xgh6bA> 8
<https://instrumentationtools.com/dp-transmitters-level-measurement> 9
<https://www.emerson.com/.../measurement-instrumentation/level>. (3 marks)

Answer: Begin with the standard definition or identity of 5 for various temperature measurement Definition of temperature - different temperature scales and conversions - simple problems. Working of bimetallic thermometer, Mercury in glass thermometer, Mercury in steel thermometer-Construction and working of radiation pyrometer, Optical pyrometer- Definition of PTC and NTC - construction and working of Resistance Temperature Detector - different types (Ni, Cu, & Pt) - characteristics of RTD - simple problems - Thermistor and its types- properties- Thermistor Characteristics- Applications of Thermistor-Seebeck effect - Peltier effect and Thomson effect - laws of thermocouple - law of intermediate temperature and law of intermediate metals - different types of industrial thermocouples- E,J,T,K,S,R - properties of different types- simple problems- thermopile - Compare the characteristics of Thermocouple, RTD and thermistor. Text / Reference: T/R Book Title/Author A.K. Sawhney, A course in Mechanical measurements and Instrumentation, R1 Dhanpat Rai R2 S.K Singh , Industrial Instrumentation and Control , TMH R3 R.K Jain , Mechanical and Industrial measurements , Khanna Publishers R4 Krishnaswami , Industrial instrumentation , New Age Publications Online Resources: Sl.No Website Link 1 <https://www.youtube.com/watch?v=4knAR4HzRjs> 2 www.temperatures.com > Vendors > Electronics > A To D 3 <https://www.youtube.com/watch?v=As5kzxkyT24> 4 <https://www.youtube.com/watch?v=K2CH1cUkMgs> 5 <https://www.youtube.com/watch?v=JHCS7ldf4aU> 6 <https://www.youtube.com/watch?v=-MQU0xgh6bA> 8 <https://instrumentationtools.com/dp-transmitters-level-measurement> 9 <https://www.emerson.com/.../measurement-instrumentation/level>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: താപനിലയുടെ നിർവ്വചനം, താപനിലയുടെ തരങ്ങൾ, താപനിലയുടെ അളവുകൾ, താപനിലയുടെ അളവുകൾ, താപനിലയുടെ അളവുകൾ.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Applying standards. Describe the construction and working.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding measurement. Compare the construction and working principle of different types of flow 8 Understanding M2.02 measurement techniques. Calibrate various flow measuring.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **principle of different types of level 8 Understanding measurement techniques. Compare and apply different transducers.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Convert different temperature scales. 3 Applying Explain the construction and working principle of different types of**

temperature 8 Understanding M4.02 measurement techniques Compare and apply different transducers Applying.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.

Term	Meaning in this lesson
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.youtube.com/watch?v=4knAR4HzRJs>
- www.temperatures.com > Vendors > Electronics > A To D
<https://www.youtube.com/watch?v=As5kzxkyT24> <https://www.youtube.com/watch?v=K2CH1cUkMgs> <https://www.youtube.com/watch?v=JHCS7ldf4aU>
[nptel.ac.in/courses/101106040/chapter 5.pdf](http://nptel.ac.in/courses/101106040/chapter%205.pdf) <https://www.youtube.com/watch?v=-MQU0xgh6bA>
- <https://instrumentationtools.com/dp-transmitters-level-measurement>
- <https://www.emerson.com/.../measurement-instrumentation/level>

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